The Brachistochrone Problem: Time Minimization via Calculus of Variations

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Abstract

We present a complete solution to the Brachistochrone problem, emphasizing both the physical intuition and the mathematical formulation. Starting from energy conservation, we derive the time functional and solve the associated variational problem using the Euler-Lagrange equation, obtaining the cycloid as the curve of fastest descent.

1 Introduction

The Brachistochrone problem is one of the most celebrated problems in the history of mathematics and physics. It was posed in 1696 by Johann Bernoulli, who challenged the leading mathematicians of Europe to determine the curve along which a particle, moving under the influence of gravity, travels between two points in the shortest time.

At first glance, one might expect the solution to be the straight line connecting the two points, since it represents the shortest distance. However, the problem reveals a deeper principle: the path of minimal time is not the path of minimal length. Instead, the solution is given by a cycloid, a curve traced by a point on the rim of a rolling circle.

The Brachistochrone problem attracted the attention of some of the greatest mathematicians of the time, including Isaac Newton, Gottfried Wilhelm Leibniz, and Jakob Bernoulli. Their solutions marked an important milestone in the development of mathematical analysis and laid the groundwork for what would later become the calculus of variations.

This problem is historically significant not only because of its elegant solution, but also because it introduced a new way of thinking about physical systems. Rather than focusing on forces alone, it emphasizes the role of variational principles, where physical trajectories are determined by the extremization of a functional. This perspective would later become central in classical mechanics, optics, and modern theoretical physics.

In this work, we present a complete solution to the Brachistochrone problem. We begin by formulating the time functional using energy conservation, and then apply the Euler–Lagrange equation to determine the curve of fastest descent. The resulting solution is shown to be a cycloid, confirming the classical result.

2 The Problem

The Brachistochrone problem consists of finding the curve along which a particle, moving under the influence of gravity, travels between two points in the least time. The key idea is that the fastest path is not the shortest one, but the one that minimizes travel time. Johann Bernoulli posed the problem of the Brachistochrone to the readers of Acta Eruditorum in June, 1696. He said:

> I, Johann Bernoulli, address the most brilliant mathematicians in the world.
>
> Nothing is more attractive to intelligent people than an honorable and challenging problem, whose possible solution will bestow fame and remain as a lasting monument. Following the example of Pascal, Fermat, and others, I hope to earn the gratitude of the scientific community by proposing a problem which will test the methods and ingenuity of those who undertake it.
>
> *Given two points A and B in a vertical plane, determine the curve traced by a point acted upon only by gravity, which starts at A and reaches B in the shortest possible time.*
>
> I shall publicly declare the name of the one who solves this problem. Solutions are to be submitted within six months.
>
> Johann Bernoulli, 1696

3 Historical Note: Newton's Solution

Among the solutions submitted to the Brachistochrone problem, that of Isaac Newton stands out for its brevity and remarkable insight. Newton did not employ the formal machinery of the calculus of variations, which had not yet been fully developed, but instead relied on a geometric and physical analogy.

His approach was based on a comparison with the propagation of light in a medium with variable density. According to Fermat's principle, light travels along the path that minimizes the time of propagation. In a non-homogeneous medium, this leads to Snell's law, where the ratio between the sine of the angle of incidence and the velocity of light remains constant.

Newton observed that a particle descending under gravity experiences a change in speed depending on its vertical position, as dictated by energy conservation. This variation of speed plays a role analogous to that of a spatially varying refractive index. By interpreting the trajectory of the particle as analogous to a light ray traversing layers of varying optical density, he concluded that the curve of fastest descent must satisfy a law similar to Snell's law.

From this reasoning, Newton was able to deduce that the required curve is a cycloid. His solution was submitted anonymously, as was customary, but Johann Bernoulli is said to have immediately recognized it, remarking that he could identify the author "*as the lion is known by his claw.*"

Newton's solution is a striking example of the power of physical intuition and analogy, anticipating the deep connections between mechanics, optics, and variational principles that would later be formalized in the calculus of variations and analytical mechanics.

4 The Solution: Infinitesimal Time Element

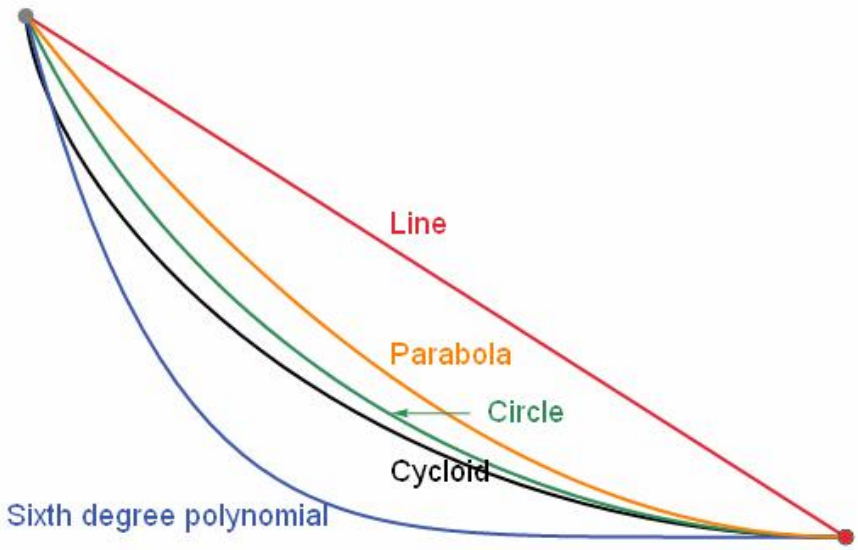

Figure 1: Brachistochrone curve with time functional $T = \int \frac{ds}{v}$.

By definition of speed,

$$v = \frac{ds}{dt}, \tag{1}$$

which implies

$$\boxed{dt = \frac{ds}{v}.} \tag{2}$$

Thus, the infinitesimal time dt is given by $dt = ds/v$.

5 Total Time Functional

The total time is obtained by integrating along the curve:

$$\boxed{T = \int dt = \int \frac{ds}{v}.} \tag{3}$$

This is the quantity we seek to minimize.

6 Expression in Terms of $y(x)$

We now express all quantities in terms of a function $y(x)$. which is the equation in the xy-coordinates. Let $y(x)$ such function, then:

The infinitesimal arc length is:

$$\boxed{ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + (y')^2} dx} \tag{4}$$

7 Why the System is Conservative

A mechanical system is said to be conservative if the forces acting on it can be derived from a scalar potential. That is,

$$\boxed{\mathbf{F} = -\nabla V.} \tag{5}$$

7.1 Gravitational Force

In the Brachistochrone problem, the particle moves under the influence of gravity:

$$\boxed{\mathbf{F} = -mg} \tag{6}$$

This force can be written as the gradient of the potential

$$\boxed{V(y) = mgy} \tag{7}$$

since

$$-\frac{dV}{dy} = -mg. \tag{8}$$

Thus, gravity is a conservative force.

7.2 Energy Conservation

The total mechanical energy is given by the Hamiltonian

$$\boxed{H = \frac{1}{2}mv^2 + V(y) = \frac{1}{2}mv^2 + mgy.} \tag{9}$$

Because the force is conservative and the system does not depend explicitly on time, the energy is conserved:

$$\boxed{\frac{dH}{dt} = 0.} \tag{10}$$

7.3 Absence of Dissipation

Another important feature is that there are no non-conservative forces such as friction or air resistance. Therefore, no energy is lost, and the total energy remains constant throughout the motion.

7.4 Conclusion

The system is conservative because the gravitational force derives from a potential and there are no dissipative forces. As a result, the total mechanical energy remains constant along the trajectory:

$$\boxed{H = E = \text{constant.}} \tag{11}$$

8 Energy Conservation and Choice of Reference

8.1 Why One Often Sets $H = 0$

In problems like the brachistochrone, one typically chooses the initial point such that

$$y = 0, \quad v = 0. \tag{12}$$

Then,

$$H = \frac{1}{2}m(0)^2 + mg(0) = 0. \tag{13}$$

Since energy is conserved,

$$H = 0 \quad \text{for all time.} \tag{14}$$

This is not a physical requirement, but simply a convenient choice of reference level for the energy.

8.2 General Case

If the particle starts at height y_0 with velocity v_0, then

$$H = \frac{1}{2}mv_0^2 + mgy_0 = E. \tag{15}$$

Along the trajectory,

$$\frac{1}{2}mv^2 + mgy = E. \tag{16}$$

Solving for the velocity,

$$\boxed{v = \sqrt{\frac{2(E - mgy)}{m}}} \tag{17}$$

9 Time Functional in Terms of General Energy

Using

$$ds = \sqrt{1 + (y')^2}dx \tag{18}$$

and the expression for the velocity from energy conservation

$$v = \sqrt{\frac{2(E - mgy)}{m}}, \tag{19}$$

thus, the total time is

$$T = \int \frac{ds}{v} = \int \frac{\sqrt{1 + (y')^2}}{\sqrt{\frac{2(E-mgy)}{m}}}dx. \tag{20}$$

This can be rewritten as

$$\boxed{T = \int \sqrt{\frac{m\left(1 + (y')^2\right)}{2(E - mgy)}}dx.} \tag{21}$$

10 Solution via Euler–Lagrange Equation

We consider the functional

$$T[y] = \int \sqrt{\frac{m\,(1+(y')^2)}{2(E-mgy)}}\,dx, \tag{22}$$

with Lagrangian

$$\boxed{L(y,y') = \sqrt{\frac{m\,(1+(y')^2)}{2(E-mgy)}}.} \tag{23}$$

The Euler-Lagrange equation is

$$\boxed{\frac{d}{dx}\left(\frac{\partial L}{\partial y'}\right) - \frac{\partial L}{\partial y} = 0.} \tag{24}$$

Since L does not depend explicitly on x, we use the Beltrami's Identity[1]

$$\boxed{L - y'\frac{\partial L}{\partial y'} = C.} \tag{10}$$

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[1] Let

$$J[y] = \int L(y,y')dx \tag{I}$$

where $L = L(y,y')$ does not depend explicitly on x. The Euler-Lagrange equation is

$$\frac{d}{dx}\left(\frac{\partial L}{\partial y'}\right) - \frac{\partial L}{\partial y} = 0 \tag{II}$$

Computing the total derivative of L with respect to x gives:

$$\frac{dL}{dx} = \frac{\partial L}{\partial y}y' + \frac{\partial L}{\partial y'}y''. \tag{III}$$

From the Euler-Lagrange equation

$$\frac{\partial L}{\partial y} = \frac{d}{dx}\left(\frac{\partial L}{\partial y'}\right). \tag{IV}$$

Substituting into the expression for dL/dx we obtain:

$$\frac{dL}{dx} = y'\frac{d}{dx}\left(\frac{\partial L}{\partial y'}\right) + \frac{\partial L}{\partial y'}y''. \tag{V}$$

We observe that

$$\frac{d}{dx}\left(y'\frac{\partial L}{\partial y'}\right) = y''\frac{\partial L}{\partial y'} + y'\frac{d}{dx}\left(\frac{\partial L}{\partial y'}\right). \tag{VI}$$

Thus

$$\frac{dL}{dx} = \frac{d}{dx}\left(y'\frac{\partial L}{\partial y'}\right). \tag{VII}$$

Rewriting

$$\frac{d}{dx}\left(L - y'\frac{\partial L}{\partial y'}\right) = 0. \tag{VIII}$$

Therefore,

$$L - y'\frac{\partial L}{\partial y'} = C \tag{IX}$$

which represents the *Beltrami's Identity.*

which is valid whenever the Lagrangian does not depend explicitly on x.

10.1 Obtaining the Differential Equation

Compute

$$\frac{\partial L}{\partial y'} = \sqrt{\frac{m}{2(E - mgy)}} \cdot \frac{y'}{\sqrt{1 + (y')^2}}. \tag{11}$$

Thus,

$$L - y'\frac{\partial L}{\partial y'} = \sqrt{\frac{m}{2(E - mgy)}} \left(\frac{1 + (y')^2 - (y')^2}{\sqrt{1 + (y')^2}} \right) \tag{12}$$

$$= \sqrt{\frac{m}{2(E - mgy)}} \cdot \frac{1}{\sqrt{1 + (y')^2}}. \tag{13}$$

Hence,

$$\boxed{\sqrt{\frac{m}{2(E - mgy)}} \cdot \frac{1}{\sqrt{1 + (y')^2}} = C.} \tag{14}$$

10.2 Algebraic Rearrangement

Squaring,

$$\frac{m}{2(E - mgy)} \cdot \frac{1}{1 + (y')^2} = C^2. \tag{15}$$

Rewriting,

$$1 + (y')^2 = \frac{m}{2C^2(E - mgy)}. \tag{16}$$

Let

$$\frac{m}{2C^2} = k, \tag{17}$$

then

$$1 + (y')^2 = \frac{k}{E - mgy}. \tag{18}$$

Thus,

$$(y')^2 = \frac{k}{E - mgy} - 1. \tag{19}$$

Thus, we obtain

$$\boxed{\frac{dx}{dy} = \sqrt{\frac{E - mgy}{k - (E - mgy)}}.} \tag{20}$$

At this point we introduce the trigonometric substitution given by:

$$\boxed{E - mgy = k \sin^2 \theta.} \tag{21}$$

i.e.

$$\boxed{y = \frac{E}{mg} - \frac{k}{mg} \sin^2 \theta} \tag{22}$$

11 Why the Trigonometric Substitution is Used

We arrive at the differential equation

$$\frac{dx}{dy} = \sqrt{\frac{E - mgy}{k - (E - mgy)}}. \tag{23}$$

11.1 Structure of the Expression

Notice that the right-hand side has the form

$$\sqrt{\frac{u}{k-u}}, \quad \text{where } u = E - mgy. \tag{24}$$

This is a standard expression that suggests a trigonometric substitution, since

$$\frac{u}{k-u} = \frac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta \tag{25}$$

if we set

$$u = k\sin^2\theta. \tag{26}$$

11.2 Simplification

With the substitution

$$E - mgy = k\sin^2\theta, \tag{27}$$

we obtain

$$k - (E - mgy) = k - k\sin^2\theta = k\cos^2\theta, \qquad \frac{E - mgy}{k - (E - mgy)} = \frac{k\sin^2\theta}{k\cos^2\theta} = \tan^2\theta. \tag{28}$$

Thus,

$$\boxed{\frac{dx}{dy} = \sqrt{\frac{E - mgy}{k - (E - mgy)}} = \tan\theta,} \tag{29}$$

which greatly simplifies the differential equation.

11.3 Geometric Insight

The substitution $u = k\sin^2\theta$ is not arbitrary: it is designed to convert the square root of a rational function into a trigonometric function. This is a standard technique for integrals involving expressions of the form $\sqrt{u/(k-u)}$, and it reveals the hidden cycloidal structure of the solution.

Then

$$y = \frac{E}{mg} - \frac{k}{mg}\sin^2\theta. \tag{30}$$

Using $\sin^2\theta = \frac{1-\cos 2\theta}{2}$, we obtain

$$\boxed{y(\theta) = \frac{E}{mg} - \frac{k}{2mg}(1 - \cos 2\theta).} \tag{31}$$

So at this point we know that

$$\boxed{\frac{dx}{dy} = \tan\theta, \quad E - mgy = k\sin^2\theta, \quad \frac{dy}{d\theta} = -\frac{2k}{mg}\sin\theta\cos\theta.} \tag{32}$$

so

$$\frac{\frac{dx}{d\theta}}{\frac{dy}{d\theta}} = \tan\theta \tag{33}$$

$$\frac{\frac{dx}{d\theta}}{-\frac{2k}{mg}\sin\theta\cos\theta} = \tan\theta \tag{34}$$

$$\begin{aligned} dx &= -\frac{2k\sin\theta\cos\theta\tan\theta d\theta}{mg} \\ &= -\frac{2k\sin^2\theta}{mg}d\theta \end{aligned} \tag{35}$$

i.e.

$$\begin{aligned} x(\theta) &= \int dx = -\frac{2k}{mg}\int d\theta\sin^2\theta \\ &= -\frac{2k}{mg}\int d\theta\left(\frac{1-\cos(2\theta)}{2}\right) \\ &= -\frac{k}{mg}\int d\theta(1-\cos(2\theta)) \\ &= -\frac{k}{mg}(\theta - \frac{\sin(2\theta)}{2}) \\ &= -\frac{k}{2mg}(2\theta - \sin(2\theta)) \end{aligned} \tag{36}$$

i.e.

$$\boxed{x(\theta) = -\frac{k}{2mg}(2\theta - \sin(2\theta))} \tag{37}$$

so we obtain

$$\boxed{y(\theta) = \frac{E}{mg} - \frac{k}{2mg}(1-\cos 2\theta), \quad x(\theta) = -\frac{k}{2mg}(2\theta - \sin(2\theta))} \tag{38}$$

using $a = \frac{k}{2mg}$, $\frac{E}{mg} = y_0$ and renaming 2θ by θ' and then again by θ the final equations can be written as:

$$\boxed{x(\theta) = a(\theta - \sin\theta), \quad y(\theta) = y_0 - a(1-\cos\theta)} \tag{39}$$

which represents a cycloid.

12 Interpretation

The problem is to find the function $y(x)$ that minimizes the functional $T[y]$. Unlike shortest path problems, here the integrand depends on both the geometry through ds and the dynamics through v.

13 Conclusion

The minimization of the time functional leads to a curve known as the cycloid. This problem is one of the earliest and most important examples in the calculus of variations, illustrating how physical principles translate into variational principles.